

Foundations of Operations Research - Notes -
v0.6.0

260236

September 2026

Preface

Every theory section in these notes has been taken from the sources:

- Course slides. [1]

About:

 [GitHub repository](#)

These notes are an unofficial resource and shouldn't replace the course material or any other book on foundations of operations research. It is not made for commercial purposes. I've made the following notes to help me improve my knowledge and maybe it can be helpful for everyone.

As I have highlighted, a student should choose the teacher's material or a book on the topic. These notes can only be a helpful material.

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Algorithms

1 Introduction

1.1 What is Operations Research?

Let's start with a definition of Operations Research (OR).

Definition 1: Operations Research (OR)

Operations Research (OR) is a branch of mathematics that uses **mathematical models** and **quantitative methods** to analyze complex decision-making problems and find optimal or near-optimal solutions.

So **Operations Research helps us make better decisions.** Suppose we have a real-world problem such as:

- Which machine should execute each job?
- How should a company allocate limited resources?
- What is the shortest route from A to B?
- How many workers should be scheduled?
- Where should a facility be built?

In all these cases, there are many possible decisions. Some are infeasible, some are feasible but bad, and some are better than others. OR tries to make this decision process **systematic and quantitative**.

In general, the idea is simple:

1. Real-world problem.
2. Formulate a mathematical representation.
3. Analysis.
4. Decision-making to find the best decision.

Example 1: Manufacturing Decision Problem

Imagine a company deciding how much of three products to manufacture. The real-world description may contain things like: limited workforce, limited machine hours, profit per product, production times.

OR translates those facts into mathematics. We might introduce variables such as:

x_1 = number of units of product 1,

x_2 = number of units of product 2,

And then construct an objective such as:

$$\max 5x_1 + 8x_2$$

Subject to resource constraints.

From the above example, we can see that OR is not simply “*doing calculations*”. A large part of OR is deciding **how to represent the decision problem mathematically in the first place**. To study decision problems, we have three major quantitative approaches:

- **Optimization** (*what should we do?*): we have decision variables, constraints, and an objective function, and we want the best feasible solution:

$$\min \text{ or } \max f(x) \quad \text{s.t. } x \in X$$

For example, minimize transportation cost while satisfying demand.

- **Game Theory** (*what should we do when others are also making strategic decisions?*): we have **multiple decision-makers** whose choices interact. The outcome for one player depends on what the others do. For example, two competing companies choosing prices.
- **Simulation** (*what will probably happen if we do this?*): we build a model of how a system behaves and then experiment with it. For example, simulate customers arriving at a call center to estimate waiting times under different staffing levels.

In these notes, we will focus on **optimization**.

🔗 Optimal vs Near-Optimal solutions

In the previous definition, we mentioned that OR finds **optimal or near-optimal solutions**. So the natural question is: “*why not always optimal?*”. If we have an optimization problem:

$$\min_{x \in X} f(x)$$

An **optimal solution** x^* satisfies:

$$f(x^*) \leq f(x) \quad \forall x \in X$$

It is literally the best feasible solution. But some **real optimization problems are computationally extremely difficult**. If the set of alternatives is enormous, finding the true optimum may **take too much time**. Then we may instead accept a solution that is:

- Feasible;
- Computationally obtainable;
- Very close to the best known solution.

That is a **near-optimal solution**. Later in the notes this distinction will become much more important, especially when we discuss combinatorial and integer optimization.

OR as an interdisciplinary field

Operations Research sits at the intersection of several areas: Applied Mathematics, Computer Science, Economics, and Industrial Engineering. Each contributes something different:

- **Applied mathematics.** Mathematics gives us the language used to describe the problem. For example:

$$\min f(x) \quad \text{s.t.} \quad g_i(x) \leq 0$$

It also gives us the theoretical tools to determine whether solutions exist, whether they are optimal, and what properties they have.

- **Computer science.** A mathematical model is not enough. Suppose a problem has 2^{100} possible solutions. Writing the problem mathematically is easy; checking all solutions is impossible in practice. We therefore need algorithms to find good solutions efficiently.

This is where computer science enters:

$$\text{Model} \longrightarrow \text{Algorithm} \longrightarrow \text{Computer implementation}$$

These notes will repeatedly follow exactly this pattern. For example, later we will see shortest-path algorithms, minimum spanning-tree algorithms, maximum-flow algorithms, Simplex, and Branch-and-Bound.

- **Economics.** Many decision problems involve quantities such as: profit, cost, scarce resources, prices, trade-offs. So economic reasoning often determines what the objective function should represent. For instance:

$$\max \text{ profit} \quad \text{or} \quad \min \text{ operating cost}$$

Later, LP duality will even give an explicit economic interpretation to certain mathematical quantities.

- **Industrial Engineering.** Industrial engineering provides many of the real problems that OR was designed to solve: scheduling, logistics, transportation, production planning, inventory, facility location, and supply chains.

So OR is not mathematics developed in isolation. It is **mathematics designed around real decision problems**.

References

- [1] Braz Pascoal Marta Margarida. Foundations of Operations Research. Slides from the HPC-E master's degree course on Politecnico di Milano, 2026.

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